Division- PTERIDOPHYTA

Sub division

Psilophytopsida	Psilotopsida	Lycopsida	Sphenopsida	Pteropsida
All are extinct plants. Plant body had only stem and rhizome. Roots and leaves were absent. Homosporous Spore tetrads were borne at the terminal sporangia. Example: *Rhynia.*	The plant body is rootless and have fungal association. Small, scaly appendages represent the leaves. Gametophyte is colourless and have fungal association. They are homosporous and spores are produced in sporangia or synangia. Example: *Psilotum.*	The plant body is differentiated into root, stem and leaves. Leaves are small, univeined, spirally arranged. Ligules are present. Sporophylls are arranged in the form of strobilus. Both homospory (*Lycopodium*) and heterospory (*Selaginella*) are found. Example: *Selaginella.*	The plant body is differentiated into root, stem and leaves. Stem shows jointed nodes and internodes. Small, scaly leaves are arranged at nodes in whorls. Peltate disc of sporangiophore possess compact strobilus. Homosporous but incipient heterospory is found in *Equisetum arvense.*	The plant body is differentiated into root, stem and leaves. Includes all the megaphyllous pteridophytes. leaf gap is present sporangia are organized into sorus. Both homosporus and heterosporus forms are present. Example: *Marsilea*

prostrate creeping system (*S. kraussiana*); suberect (*S. rupestris*); erect (*Selaginella erythropus*); climbing (*Selaginella alligans*). *S. oregana* is an epiphyte. Most of the species are perennials. On the basis of structure of stem and arrangement of leaves, *Selaginella* is divided into two sub genera namely Homoeophyllum and Heterophyllum.



The success and dominance of vascular plants is due to the development of

- Extensive root system.
- Efficient conducting tissues.
- Cuticle to prevent desiccation.
- Stomata for effective gaseous exchange.

Homeophyllum include species with erect stem and spirally arranged leaves. (Example: *S. upestris* and *S. oregana*). Heterophyllum include species with prosrate stem with short erect branches and dimorphic leaves (Example: *S. kraussiana* and *S. lepidophylla*).

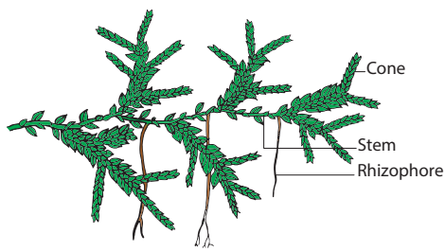


Figure 2.16: *Selaginella* Habit

Root

Primary roots are short lived and the adult plant has adventitious roots. The root may arise at the point of dichotomous branching or knot like swelling present at the basal portion of the stem. Roots are endogenous in origin.

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